

# Koran

50 topics · 139 questions. Each day 10 are picked at random. True/false has one try; other questions have two. Any topic missed comes back the next day with a different question. Correct answers are in green (the dashboard shuffles the answer order).

## Anatomy & physiology 6 topics

### Anatomy vs. physiology

1. MULTIPLE CHOICE

**Anatomy is the study of:**

✓ **Body structure and the physical relationships between body parts**

How the body functions

Only diseases of the body

Only the cells of the body

2. TRUE OR FALSE

**Physiology is the study of how the body functions.**

Answer: **True**

3. ANALYZE THE CASE

**A student measures how fast the heart pumps blood during exercise. Is the student studying anatomy or physiology?**

✓ **Physiology**

Anatomy

Cytology

Surface anatomy

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**Explanation shown if missed:** Anatomy = STRUCTURE (what a body part is and where it is). Physiology = FUNCTION (how it works). Studying the shape of the heart is anatomy; studying how the heart pumps is physiology.

### Gross vs. microscopic anatomy

1. MULTIPLE CHOICE

**Which area of anatomy examines structures visible to the naked eye?**

✓ **Gross (macroscopic) anatomy**

Microscopic anatomy

Cytology

Histology

2. ANALYZE THE CASE

**A scientist studies a thin slice of the stomach wall under a microscope. Which area of anatomy is this?**

✓ **Microscopic anatomy**

Gross anatomy

Surface anatomy

Regional anatomy

3. TRUE OR FALSE

**Microscopic anatomy studies structures that are visible to the naked eye.**

Answer: **False**

**Explanation shown if missed:** Gross (macroscopic) anatomy looks at things you can see with your eyes, like the heart or a bone. Microscopic anatomy looks at things too small to see without a microscope.

## Cytology & histology

1. MULTIPLE CHOICE

**Histology is the study of:**

✓ **Tissues**

Cells

Organ systems

Body regions

2. TRUE OR FALSE

**Cytology is the study of cells.**

Answer: **True**

3. ANALYZE THE CASE

**A lab looks at individual cells scraped from the inside of the cheek. Which field is this?**

✓ **Cytology**

Histology

Regional anatomy

Systemic physiology

**Explanation shown if missed:** Cytology = the study of CELLS (think "cyto" = cell). Histology = the study of TISSUES, which are groups of cells working together.

## Surface, regional & systemic anatomy

1. MULTIPLE CHOICE

**Which area of anatomy studies general form and superficial markings?**

✓ **Surface anatomy**

Regional anatomy

Microscopic anatomy

Developmental anatomy

2. ANALYZE THE CASE

**A textbook chapter covers every structure in the head and neck together. Which area of anatomy is it using?**

✓ **Regional anatomy**

Systemic anatomy

Cytology

Developmental anatomy

3. ANALYZE THE CASE

**A class studies the whole digestive system, from mouth to rectum. Which area of anatomy is this?**

✓ **Systemic anatomy**

Regional anatomy

Surface anatomy

Histology

**Explanation shown if missed:** Surface anatomy = outside form and markings. Regional anatomy = everything in one body AREA (like the head and neck). Systemic anatomy = one organ SYSTEM at a time (like the digestive system).

## Developmental & clinical anatomy

1. MULTIPLE CHOICE

**Which area of anatomy studies changes in form from conception through physical maturity?**

✓ **Developmental anatomy**

Clinical anatomy

Surface anatomy

Systemic anatomy

2. ANALYZE THE CASE

**A doctor uses knowledge of anatomy to find the cause of a patient's pain and plan treatment. Which area is this?**

✓ **Clinical (pathological) anatomy**

Developmental anatomy

Cytology

Surface anatomy

**Explanation shown if missed:** Developmental anatomy follows how the body's form changes from conception until physical maturity. Clinical (pathological) anatomy uses anatomy to diagnose and treat patients.

## Areas of physiology

1.

**Which area of physiology studies chemical processes within and between cells?**

✓ **Cell physiology**

Systemic physiology

Special (organ) physiology

Clinical physiology

2.

**A researcher studies how diabetes damages the kidneys. Which area of physiology is this?**

✓ **Clinical (pathological) physiology**

Cell physiology

Special (organ) physiology

Systemic physiology

3.

**Special (organ) physiology studies the functions of specific organs.**

Answer: **True**

4.

**Systemic physiology studies the functions of an organ system.**

Answer: **True**

5.

**A scientist studies how the lungs, airways, and diaphragm work together as a whole system to move air. Which area of physiology is this?**

✓ **Systemic physiology**

Cell physiology

Clinical (pathological) physiology

Microscopic anatomy

**Explanation shown if missed:** Cell physiology = chemical processes in and between cells. Special (organ) physiology = one specific organ. Systemic physiology = a whole organ system. Clinical (pathological) physiology = how disease affects organs or systems.

## Organization & homeostasis 14 topics

### Levels of organization

1. MULTIPLE CHOICE

**Which list shows the levels of organization in the correct order, from smallest to largest?**

✓ **Atoms → Molecules → Organelles → Cells → Tissues → Organs → Organ systems → Organism**

Cells → Atoms → Molecules → Tissues → Organelles → Organs → Organ systems → Organism

Molecules → Atoms → Cells → Organelles → Organs → Tissues → Organism → Organ systems

Atoms → Molecules → Cells → Organelles → Tissues → Organ systems → Organs → Organism

2. MULTIPLE CHOICE

**Which level comes right after TISSUES?**

✓ **Organs**

Cells

Organ systems

Organelles

3. TRUE OR FALSE

**An organ is composed of tissues.**

Answer: **True**

4. MULTIPLE CHOICE

**Tissues are composed of:**

✓ **Cells**

Organs

Organ systems

Organisms

---

**Explanation shown if missed:** From smallest to largest: Atoms → Molecules → Organelles → Cells → Tissues → Organs → Organ systems → Organism. Small pieces build bigger pieces, like letters → words → sentences → a story.

## Atom vs. cell

1. MULTIPLE CHOICE

**What is the smallest LIVING unit of the body?**

✓ **Cell**

Atom

Molecule

Organelle

2. TRUE OR FALSE

**The atom is the smallest living unit of the body.**

Answer: **False**

---

**Explanation shown if missed:** The atom is the smallest stable unit of matter, but it is NOT alive. The cell is the smallest LIVING unit.

## Organ system

1. MULTIPLE CHOICE

**An organ system is:**

✓ **A group of organs with a unique collective function**

- A group of similar cells
- The smallest living unit
- A single organ

2. ANALYZE THE CASE

**The mouth, stomach, and intestines work together to break down food. This group is an example of a(n):**

✓ **Organ system**

- Tissue
- Organelle
- Molecule

---

**Explanation shown if missed:** An organ system is a group of organs that work together for one shared job. Example: the heart, blood vessels, and blood work together to move blood around the body.

## The 11 organ systems

1. SELECT ALL

**Select the organ systems that are part of the 11 organ systems of the body.**

✓ **Integumentary**

✓ **Lymphatic**

✓ **Endocrine**

✓ **Urinary**

- Mediastinal
- Peritoneal

2. MULTIPLE CHOICE

**How many organ systems does the human body have?**

✓ **11**

- 7
- 4
- 20

3. TRUE OR FALSE

**The integumentary, skeletal, muscular, and nervous systems are 4 of the 11 organ systems.**

Answer: **True**

---

**Explanation shown if missed:** The body has 11 organ systems: integumentary, skeletal, muscular, nervous, endocrine, cardiovascular, lymphatic, respiratory, digestive, urinary, and reproductive.

## Integumentary system

1. MULTIPLE CHOICE

**What is the main function of the integumentary system?**

✓ **Protection**

Movement

Controlling immediate responses

Producing blood cells

2. ANALYZE THE CASE

**Your skin keeps germs and harmful things out of your body. Which organ system is doing this job?**

✓ **Integumentary**

Skeletal

Muscular

Nervous

3. TRUE OR FALSE

**The main function of the integumentary system is movement.**

Answer: **False**

---

**Explanation shown if missed:** The integumentary system is the skin (plus hair and nails). Its main job is PROTECTION, like a raincoat that covers the whole body and keeps germs out.

## Skeletal system

1. MULTIPLE CHOICE

**Which organ system provides support and protection for the body?**

✓ **Skeletal**

Muscular

Nervous

Digestive

2. ANALYZE THE CASE

**The skull surrounds the brain and the ribs surround the heart and lungs. Which organ system is protecting these organs?**

✓ **Skeletal**

Muscular

Nervous

Digestive

3. TRUE OR FALSE

**The skeletal system gives the body support and protection.**

Answer: **True**

**Explanation shown if missed:** The skeletal system (bones) gives the body SUPPORT, like the frame of a house, and PROTECTION, like the skull around the brain and the ribs around the heart and lungs.

## Muscular system

1. MULTIPLE CHOICE

**What is the main function of the muscular system?**

✓ **Movement**

- Protection of the body surface
- Controlling immediate responses
- Producing hormones

2. ANALYZE THE CASE

**You run to catch the bus. Which organ system produces the movement of your legs?**

✓ **Muscular**

- Integumentary
- Endocrine
- Urinary

3. TRUE OR FALSE

**The muscular system is responsible for movement.**

Answer: **True**

**Explanation shown if missed:** The muscular system is in charge of MOVEMENT. Muscles pull on bones so you can walk, run, and even smile.

## Nervous system (organ system)

1. MULTIPLE CHOICE

**Which organ system controls immediate responses?**

✓ **Nervous**

- Muscular
- Skeletal
- Integumentary

2. ANALYZE THE CASE

**A ball flies toward your face and you blink instantly. Which organ system controlled this immediate response?**

✓ **Nervous**

- Integumentary
- Skeletal
- Digestive

3. TRUE OR FALSE

**The nervous system's main job is to support and protect the body like a frame.**

Answer: **False**

**Explanation shown if missed:** The nervous system CONTROLS IMMEDIATE RESPONSES. It reacts in a split second, like blinking when something flies toward your eye.

## Organ systems work together

1. MULTIPLE CHOICE

**All organ systems depend on each other (are interdependent) in order to preserve a stable internal environment, also known as:**

✓ **Homeostasis**

Positive feedback

Anatomical position

Metabolism

2. TRUE OR FALSE

**Each organ system works alone and does not depend on the other systems.**

Answer: **False**

3. ANALYZE THE CASE

**During exercise, your breathing and heart rate both speed up to keep oxygen in your blood stable. What does this show?**

✓ **Organ systems work together to maintain homeostasis**

Organ systems work independently

Only one organ system is working

The body is using anatomy, not physiology

**Explanation shown if missed:** All organ systems depend on each other (they are interdependent) to keep the inside of the body stable. That stable inside environment is called HOMEOSTASIS.

## Characteristics of life

1. SELECT ALL

**Select the characteristics of life listed in the review guide.**

✓ **Cells**

✓ **Metabolism**

✓ **Reproduction**

✓ **Response to the environment**

Photosynthesis

Living only in water

2. MULTIPLE CHOICE

**In the memory trick CHRMD, what does the M stand for?**

✓ **Metabolism**

Movement

Muscles

Membranes

3. ANALYZE THE CASE

**You touch a hot pan and pull your hand away. Which characteristic of life does this show?**

✓ **Response to the environment**

Reproduction

Adaptation over generations

Heredity

**Explanation shown if missed:** Living things have cells, metabolism, reproduction, response to the environment, and adaptation (evolution). The slides' memory trick is CHRMD: Cells, Homeostasis, Reproduction, Metabolism, DNA/Heredity.

## Homeostasis

1. MULTIPLE CHOICE

**Homeostasis is:**

✓ **Maintenance of a relatively stable internal environment**

The production of new cells

The study of tissues

An increase in body size

2. ANALYZE THE CASE

**On a hot day you start to sweat and your body temperature stays near normal. What is your body doing?**

✓ **Maintaining homeostasis**

Using positive feedback

Reproducing

Changing its anatomy

3. TRUE OR FALSE

**Homeostasis means the internal environment changes wildly all the time.**

Answer: **False**

**Explanation shown if missed:** Homeostasis means keeping the inside of your body relatively stable, even when things outside change. Example: sweating on a hot day keeps your body temperature near normal.

## Negative feedback

1. MULTIPLE CHOICE

**In negative feedback, the response:**

✓ **Opposes/reduces the original stimulus**

Enhances/amplifies the original stimulus

Has no effect on the stimulus

Stops all body functions

2. ANALYZE THE CASE

**After a meal your blood glucose rises; insulin is released and glucose drops back to normal. This is an example of:**

✓ **Negative feedback**

Positive feedback

Cardiac tamponade

Developmental anatomy

3. TRUE OR FALSE

**Control of body temperature is an example of negative feedback.**

Answer: **True**

**Explanation shown if missed:** In negative feedback, the body's response goes the OPPOSITE way of the change, bringing things back to normal. Examples: body temperature, blood pressure, blood glucose, and blood calcium.

## Positive feedback

1. MULTIPLE CHOICE

**In positive feedback, the response:**

✓ **Enhances/amplifies the original stimulus**

Opposes/reduces the original stimulus

Always restores the normal set point right away

Only happens in plants

2. ANALYZE THE CASE

**During childbirth, contractions trigger a hormone that makes the contractions even stronger. This is:**

✓ **Positive feedback**

Negative feedback

Homeostatic set point

A sectional plane

3. **SELECT ALL**

**Select the examples of POSITIVE feedback.**

✓ **Childbirth**

✓ **Blood clotting**

✓ **Lactation**

Body temperature regulation

Blood pressure regulation

4. **TRUE OR FALSE**

**In this course, coughing and sneezing are examples of positive feedback.**

Answer: **True**

5. **SELECT ALL**

**Select the examples of positive feedback listed in the course.**

✓ **Coughing**

✓ **Sneezing**

✓ **Breastfeeding (lactation)**

Blood glucose regulation

Blood calcium regulation

**Explanation shown if missed:** In positive feedback, the response makes the change even BIGGER until an event is finished. Examples from the course: childbirth (contractions get stronger), blood clotting, lactation (breastfeeding), coughing, and sneezing.

## Receptor → control center → effector

1. **MULTIPLE CHOICE**

**What is the correct order of a homeostatic regulatory mechanism?**

✓ **Receptor → Control center → Effector**

Effector → Receptor → Control center

Control center → Effector → Receptor

Receptor → Effector → Control center

2. **ANALYZE THE CASE**

**Skin sensors detect cold, the brain processes it, and muscles start to shiver. In this example, the muscles are the:**

✓ **Effector**

Receptor

Control center

Stimulus

3. **TRUE OR FALSE**

**The control center is the part that senses a change.**

Answer: **False**

**Explanation shown if missed:** Receptor = senses a change. Control center = processes the information and sends instructions. Effector = carries out the instructions. Example: skin senses cold →

brain decides → muscles shiver.

## Position, directions & planes 11 topics

### Anatomical position

1. MULTIPLE CHOICE

**Which describes the anatomical position?**

✓ **Standing erect, facing forward, arms at the sides, palms forward, feet slightly apart**

Lying face down with arms over the head

Standing with palms facing backward and feet crossed

Sitting with arms folded

2. TRUE OR FALSE

**In anatomical position, the palms face backward.**

Answer: **False**

**Explanation shown if missed:** Anatomical position: standing up straight, facing forward, arms at the sides, palms facing FORWARD, feet slightly apart. Doctors use it so everyone describes the body the same way.

### Supine vs. prone

1. MULTIPLE CHOICE

**A patient lying on their back with the face upward is in the \_\_\_ position.**

✓ **Supine**

Prone

Anatomical

Lateral

2. ANALYZE THE CASE

**For an exam of the back, the patient lies on their belly with the face down. This position is:**

✓ **Prone**

Supine

Anatomical

Superior

3. TRUE OR FALSE

**Prone means lying face down.**

Answer: **True**

**Explanation shown if missed:** Supine = lying on your back, face UP (you could hold a bowl of soup: "sup"). Prone = lying on your belly, face DOWN.

## Superior vs. inferior

1. MULTIPLE CHOICE

The head is \_\_\_ to the chest.

✓ Superior

Inferior

Distal

Deep

2. TRUE OR FALSE

Inferior means toward the feet.

Answer: True

3. ANALYZE THE CASE

The knee is closer to the feet than the hip is. The knee is \_\_\_ to the hip.

✓ Inferior

Superior

Medial

Superficial

---

**Explanation shown if missed:** Superior = toward the HEAD (higher). Inferior = toward the FEET (lower). The head is superior to the chest.

## Anterior vs. posterior

1. MULTIPLE CHOICE

The sternum (breastbone) is \_\_\_ to the spine.

✓ Anterior (ventral)

Posterior (dorsal)

Lateral

Proximal

2. TRUE OR FALSE

Dorsal means the same as posterior.

Answer: True

3. ANALYZE THE CASE

The shoulder blades are on the back of the body and the belly button is on the front. The shoulder blades are \_\_\_ to the belly button.

✓ Posterior (dorsal)

Anterior (ventral)

Distal

Medial

---

**Explanation shown if missed:** Anterior (ventral) = toward the FRONT. Posterior (dorsal) = toward the BACK. The breastbone is anterior to the spine.

## Medial vs. lateral

1. MULTIPLE CHOICE

The nose is \_\_\_ to the eyes.

✓ Medial

Lateral

Distal

Superficial

2. ANALYZE THE CASE

The ears are on the sides of the head, far from the midline. The ears are \_\_\_ to the nose.

✓ Lateral

Medial

Proximal

Deep

3. TRUE OR FALSE

Lateral means toward the midline of the body.

Answer: **False**

---

**Explanation shown if missed:** Medial = toward the MIDLINE (an imaginary line down the middle of the body). Lateral = AWAY from the midline, toward the sides.

## Proximal vs. distal

1. MULTIPLE CHOICE

The elbow is \_\_\_ to the wrist.

✓ Proximal

Distal

Lateral

Inferior

2. ANALYZE THE CASE

The fingers are farther than the elbow from where the arm attaches to the body. The fingers are \_\_\_ to the elbow.

✓ Distal

Proximal

Medial

Superior

---

**Explanation shown if missed:** Proximal = CLOSER to where a limb attaches to the body. Distal = FARTHER from that point. The elbow is proximal to the wrist; the fingers are distal to the elbow.

## Superficial vs. deep

1. MULTIPLE CHOICE

The skin is \_\_\_ to the muscles.

✓ **Superficial**

Deep

Proximal

Caudal

2. TRUE OR FALSE

**Deep means farther from the surface of the body.**

Answer: **True**

---

**Explanation shown if missed:** Superficial = closer to the body SURFACE. Deep = farther INSIDE, away from the surface. The skin is superficial to the muscles.

## Cranial vs. caudal

1. MULTIPLE CHOICE

**Caudal means:**

✓ **Toward the lower end or tail**

Toward the head

Toward the front

Toward the midline

2. TRUE OR FALSE

**Cranial means toward the head.**

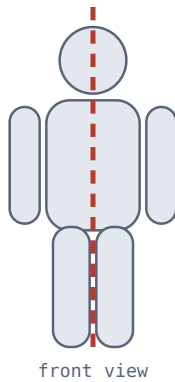
Answer: **True**

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**Explanation shown if missed:** Cranial = toward the HEAD (cranium = skull). Caudal = toward the lower end or TAIL ("cauda" means tail).

## Sagittal plane

1. MULTIPLE CHOICE



**Which plane is shown by the dashed line?**

✓ **Sagittal**

Coronal (frontal)

Transverse (horizontal)

Diagonal

2. MULTIPLE CHOICE

**Which plane divides the body into left and right portions?**

✓ **Sagittal**

Coronal (frontal)

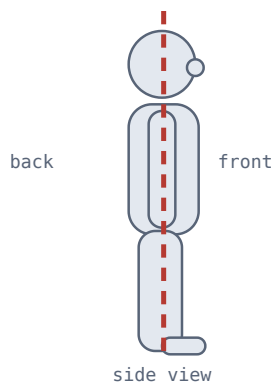
Transverse (horizontal)

Superficial

**Explanation shown if missed:** A sagittal plane cuts the body into LEFT and RIGHT portions. Memory trick: Sagittal = Sides.

## Coronal (frontal) plane

1. MULTIPLE CHOICE



**In this side view, which plane is shown by the dashed line?**

✓ **Coronal (frontal)**

Sagittal

Transverse (horizontal)

Diagonal

2. ANALYZE THE CASE

**A cut separates the face and chest from the back of the body. Which plane made this cut?**

✓ **Coronal (frontal)**

Sagittal

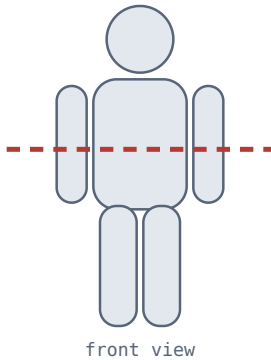
Transverse (horizontal)

Midline

**Explanation shown if missed:** A coronal (frontal) plane cuts the body into a FRONT (anterior) piece and a BACK (posterior) piece. Memory trick: Coronal = front/back.

## Transverse (horizontal) plane

1. MULTIPLE CHOICE



**Which plane is shown by the dashed line?**

✓ **Transverse (horizontal)**

Sagittal

Coronal (frontal)

Vertical

2. ANALYZE THE CASE

**A CT scan shows "slices" that separate the body into upper and lower parts. Which plane are these slices in?**

✓ **Transverse (horizontal)**

Sagittal

Coronal (frontal)

Oblique

**Explanation shown if missed:** A transverse (horizontal) plane cuts the body into a TOP (superior) piece and a BOTTOM (inferior) piece. Memory trick: Transverse = top/bottom, like slicing bread.

## Body cavities & membranes 6 topics

### Ventral body cavity

1. MULTIPLE CHOICE

**The ventral body cavity includes the:**

✓ **Thoracic and abdominopelvic cavities**

Cranial and spinal cavities

Cranial cavity only

Pericardial and cranial cavities

2. TRUE OR FALSE

**The diaphragm separates the thoracic cavity from the abdominopelvic cavity.**

Answer: **True**

3. ANALYZE THE CASE

**A wound tears through the diaphragm. Which two cavities are no longer separated?**

✓ **Thoracic and abdominopelvic**

Cranial and spinal

Pericardial and pleural

Spinal and abdominopelvic

**Explanation shown if missed:** The ventral (front) cavity has two parts: the thoracic cavity (chest) and the abdominopelvic cavity (belly and pelvis). The diaphragm muscle separates them.

## Dorsal body cavity

1. MULTIPLE CHOICE

**The dorsal body cavity contains the:**

✓ **Cranial cavity (brain) and spinal cavity (spinal cord)**

Heart and lungs

Stomach and liver

Kidneys and bladder

2. TRUE OR FALSE

**The brain is located in the ventral body cavity.**

Answer: **False**

**Explanation shown if missed:** The dorsal (back) cavity has the cranial cavity, which holds the BRAIN, and the spinal cavity, which holds the SPINAL CORD.

## Thoracic cavity

1. MULTIPLE CHOICE

**The heart sits in the \_\_\_ cavity, and each lung sits in a \_\_\_ cavity.**

✓ **pericardial; pleural**

pleural; pericardial

cranial; spinal

abdominal; pelvic

2. TRUE OR FALSE

**The thoracic cavity contains the heart and lungs.**

Answer: **True**

**Explanation shown if missed:** The thoracic cavity (chest) holds the heart and lungs. The heart sits in the PERICARDIAL cavity; each lung sits in its own PLEURAL cavity.

## Mediastinum

1. MULTIPLE CHOICE

**The mediastinum is:**

✓ **The central compartment of the thoracic cavity**

The space around each lung

The lining of the abdominopelvic cavity

Part of the dorsal body cavity

2. SELECT ALL

**Select the structures found in the mediastinum.**

✓ **Heart**

✓ **Esophagus**

✓ **Trachea**

✓ **Thymus**

Lungs

Kidneys

3. TRUE OR FALSE

**The lungs are located inside the mediastinum.**

Answer: **False**

**Explanation shown if missed:** The mediastinum is the central compartment of the chest, between the lungs. It holds the heart, great vessels, esophagus, trachea, phrenic and cardiac nerves, thoracic duct, thymus, and lymph nodes. The lungs are NOT in it.

## Serous membranes

1. MULTIPLE CHOICE

**Which serous membrane is associated with the lungs?**

✓ **Pleurae**

Pericardium

Peritoneum

Mediastinum

2. MULTIPLE CHOICE

**Which serous membrane is associated with the abdominopelvic cavity?**

✓ **Peritoneum**

Pleurae

Pericardium

Diaphragm

3. ANALYZE THE CASE

**Each time the heart beats, it rubs against the tissue around it. What reduces this friction?**

✓ **Serous fluid**

Blood in the dural sinuses

Air in the lungs

The diaphragm

**Explanation shown if missed:** Serous membranes are thin linings: PERICARDIUM = heart, PLEURAE = lungs, PERITONEUM = abdominopelvic cavity. Their serous fluid lets organs move and expand without friction.

## Parietal vs. visceral layers

1. MULTIPLE CHOICE

**The visceral layer of a serous membrane:**

✓ **Lines/covers the internal organ**

Lines the cavity/body wall

Is filled with air

Is part of the skin

2. ANALYZE THE CASE

**The layer of pleura attached directly to the surface of the lung is the:**

✓ **Visceral pleura**

Parietal pleura

Pericardium

Peritoneum

3. TRUE OR FALSE

**The parietal layer lines the cavity or body wall.**

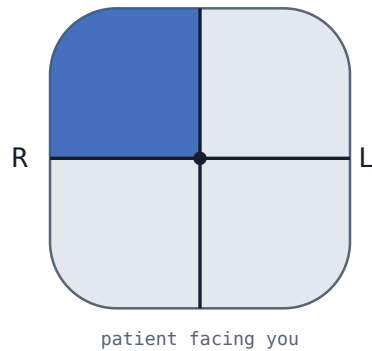
Answer: **True**

**Explanation shown if missed:** Picture pushing your fist into a balloon: the layer touching your fist is VISCERAL (covers the organ); the outer layer is PARIETAL (lines the cavity or body wall).

## Abdominal regions & organs 5 topics

### Abdominal quadrants

1. MULTIPLE CHOICE



Which quadrant is highlighted? (R = patient's right)

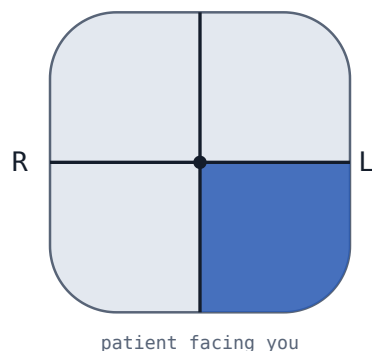
✓ **RUQ — Right Upper Quadrant**

LUQ — Left Upper Quadrant

RLQ — Right Lower Quadrant

LLQ — Left Lower Quadrant

2. MULTIPLE CHOICE



Which quadrant is highlighted? (R = patient's right)

✓ **LLQ — Left Lower Quadrant**

RLQ — Right Lower Quadrant

LUQ — Left Upper Quadrant

RUQ — Right Upper Quadrant

3. MULTIPLE CHOICE

What does LUQ stand for?

✓ **Left Upper Quadrant**

Lower Upper Quadrant

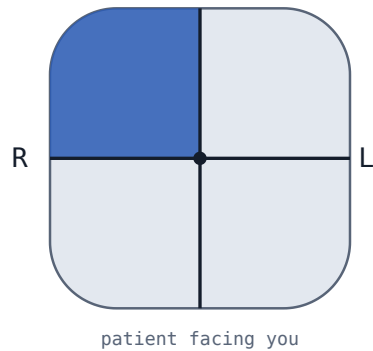
Lateral Upper Quadrant

Left Under Quadrant

**Explanation shown if missed:** The abdomen is split into 4 quadrants by a vertical and a horizontal line through the belly button: RUQ, LUQ, RLQ, LLQ. Right and left are always the PATIENT's right and left.

## Organs in the abdominal quadrants

1. MULTIPLE CHOICE



**Which organ is found in the highlighted quadrant? (R = patient's right)**

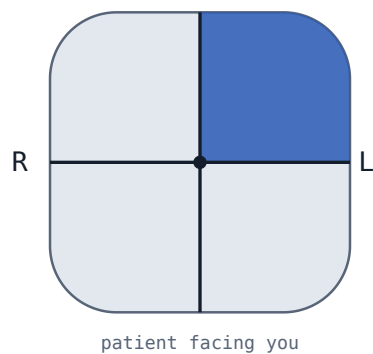
✓ **Liver**

Appendix

Stomach

Urinary bladder

2. MULTIPLE CHOICE



**Which organ is found in the highlighted quadrant? (R = patient's right)**

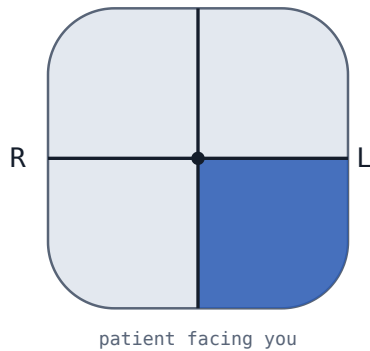
✓ **Stomach**

Liver

Appendix

Urinary bladder

3. MULTIPLE CHOICE



**Which organ is found in the highlighted quadrant? (R = patient's right)**

✓ **Large intestine**

Liver

Appendix

Stomach

4. MULTIPLE CHOICE

**The appendix is located in the:**

✓ **RLQ — Right Lower Quadrant**

LUQ — Left Upper Quadrant

RUQ — Right Upper Quadrant

LLQ — Left Lower Quadrant

5. ANALYZE THE CASE

**A patient has sharp pain in the RLQ, and the doctor suspects appendicitis. Why does the location make sense?**

✓ **The appendix is in the right lower quadrant**

The liver is in the right lower quadrant

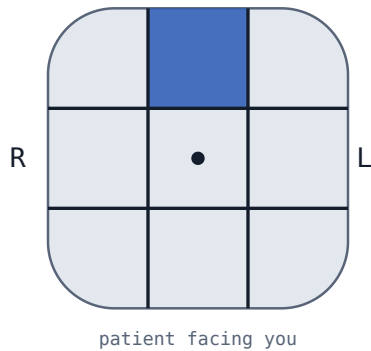
The stomach is in the right lower quadrant

The appendix is in the left upper quadrant

**Explanation shown if missed:** One organ in each quadrant: RUQ = liver, LUQ = stomach, RLQ = appendix, LLQ = large intestine. Right and left are always the PATIENT's right and left.

## Epigastric, umbilical & hypogastric regions

1. MULTIPLE CHOICE



Which of the 9 abdominal regions is highlighted?

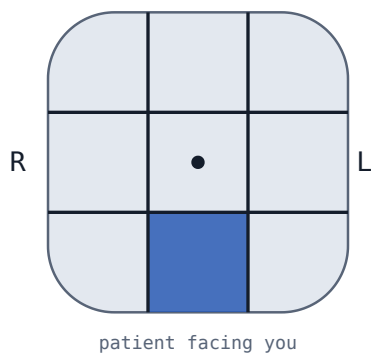
✓ **Epigastric**

Umbilical

Hypogastric

Right hypochondriac

2. MULTIPLE CHOICE



Which of the 9 abdominal regions is highlighted?

✓ **Hypogastric**

Umbilical

Epigastric

Left iliac (inguinal)

3. MULTIPLE CHOICE

Which region is in the CENTER of the 9 abdominal regions?

✓ **Umbilical**

Epigastric

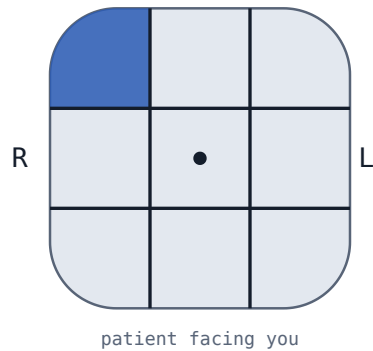
Hypogastric

Left lumbar

**Explanation shown if missed:** The middle column of the 9 regions, from top to bottom: EPIGASTRIC (above the stomach area), UMBILICAL (around the belly button), HYPOGASTRIC (below the belly button).

**Hypochondriac, lumbar & iliac regions**

1. MULTIPLE CHOICE



Which of the 9 abdominal regions is highlighted? (R = patient's right)

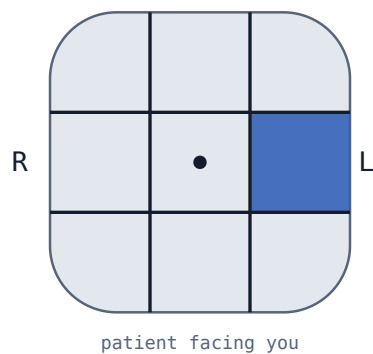
✓ **Right hypochondriac**

Left hypochondriac

Epigastric

Right lumbar

2. MULTIPLE CHOICE



Which of the 9 abdominal regions is highlighted? (R = patient's right)

✓ **Left lumbar**

Right lumbar

Umbilical

Left iliac (inguinal)

3. MULTIPLE CHOICE

Which regions make up the **BOTTOM** row of the 9 abdominal regions?

✓ **Right iliac, hypogastric, left iliac**

Right lumbar, umbilical, left lumbar

Right hypochondriac, epigastric, left hypochondriac

Right iliac, umbilical, left lumbar

**Explanation shown if missed:** The side columns, from top to bottom: HYPOCHONDRIAC (under the ribs), LUMBAR (beside the belly button), ILIAC/INGUINAL (near the hips and groin). Each has a right and a left.

## Retroperitoneal organs (SAD PUCKER)

1.

**What does retroperitoneal mean?**

✓ **Situated behind the peritoneum**

Inside the pericardium

Above the diaphragm

Inside the cranial cavity

2.

**Select the retroperitoneal organs (SAD PUCKER).**

✓ **Kidneys**

✓ **Pancreas**

✓ **Abdominal aorta**

✓ **Ureters**

Brain

Lungs

3.

**In the memory trick SAD PUCKER, what does the K stand for?**

✓ **Kidneys**

Knees

Keratin

Kinetic energy

---

**Explanation shown if missed:** Retroperitoneal = BEHIND the peritoneum. Remember SAD PUCKER: Suprarenal glands, Abdominal aorta, Duodenum, Pancreas, Ureters, Colon (ascending/descending), Kidneys, Esophagus (inferior part), Rectum.

## Clinical conditions & imaging 8 topics

### Pleuritis & pleural effusion

1.

**Pleuritis (pleurisy) is:**

✓ **Inflammation of the pleurae**

Fluid in the peritoneal cavity

Compression of the heart

Inflammation of the peritoneum

2. ANALYZE THE CASE

**A chest X-ray shows abnormal fluid collecting in the spaces around the lungs. This condition is:**

✓ **Pleural effusion**

- Ascites
- Pericarditis
- Peritonitis

**Explanation shown if missed:** Pleuritis (pleurisy) = the pleurae around the lungs are INFLAMED.  
Pleural effusion = extra FLUID builds up in the spaces around the lungs.

## Pericarditis & cardiac tamponade

1. MULTIPLE CHOICE

**Cardiac tamponade is:**

✓ **Compression of the heart from blood or fluid in the pericardial sac**

- Inflammation of the pleurae
- Fluid buildup in the peritoneal cavity
- A tear in the diaphragm

2. TRUE OR FALSE

**Pericarditis is inflammation of the pericardium.**

Answer: **True**

3. ANALYZE THE CASE

**After a car accident, blood collects around the heart inside its sac and squeezes it. This is:**

✓ **Cardiac tamponade**

- Pleurisy
- Ascites
- Peritonitis

**Explanation shown if missed:** Pericarditis = the pericardium (sac around the heart) is inflamed.  
Cardiac tamponade = blood or fluid fills the pericardial sac and SQUEEZES the heart so it can't pump well.

## Peritonitis & ascites

1. MULTIPLE CHOICE

**Ascites is:**

✓ **Abnormal accumulation of fluid in the peritoneal cavity**

- Inflammation of the peritoneum
- Fluid around the lungs
- Inflammation of the heart sac

2. TRUE OR FALSE

**Peritonitis is inflammation of the peritoneum.**

Answer: **True**

**Explanation shown if missed:** Peritonitis = the peritoneum (lining of the abdominopelvic cavity) is INFLAMED. Ascites = extra FLUID builds up in the peritoneal cavity, making the belly swell.

## X-ray & CT

1. MULTIPLE CHOICE

**Which imaging method combines X-ray images from different angles to create cross-sectional images?**

✓ **CT**

MRI

Ultrasound

PET

2. ANALYZE THE CASE

**A child falls and the doctor wants a quick look at a possible broken arm bone. Which imaging is most useful for bone?**

✓ **X-ray**

PET

Angiography

Nuclear scan

**Explanation shown if missed:** X-ray uses high-energy radiation that passes through the body; it is great for bones. CT combines many X-ray images from different angles to make cross-section "slices."

## MRI

1. MULTIPLE CHOICE

**Which imaging procedure uses radio waves?**

✓ **MRI**

CT

X-ray

Angiography

2. ANALYZE THE CASE

**A doctor needs a very detailed image of the soft tissues inside a knee. Which imaging method fits best?**

✓ **MRI**

X-ray

Nuclear scan

Angiography

3. TRUE OR FALSE

**MRI uses high-energy X-ray radiation.**

Answer: **False**

**Explanation shown if missed:** MRI uses pulses of RADIO WAVES (with a strong magnet) and gives very detailed pictures of soft tissue. It is the imaging method in this chapter that uses radio waves.

## Ultrasound

1. MULTIPLE CHOICE

**Ultrasound creates images using:**

✓ **High-frequency sound waves and echoes**

Pulses of radio waves

A radioactive tracer

X-rays from different angles

2. ANALYZE THE CASE

**A pregnant patient has an imaging test that bounces sound waves off the baby to make a picture. Which test is this?**

✓ **Ultrasound**

CT

PET

X-ray

**Explanation shown if missed:** Ultrasound sends high-frequency SOUND waves into the body and uses the echoes to create images, like a bat using echoes to "see."

## Nuclear scan & PET

1. MULTIPLE CHOICE

**A PET scan assesses:**

✓ **Metabolic/physiological activity using a tracer**

Bone structure using sound waves

Blood vessels using a catheter only

Soft tissue using radio waves

2. TRUE OR FALSE

**A nuclear scan uses a radioactive tracer.**

Answer: **True**

**Explanation shown if missed:** A nuclear scan uses a radioactive TRACER (radiopharmaceutical). A PET scan also uses a tracer and shows how active tissues are (their metabolism).

## Angiography

1. **MULTIPLE CHOICE**

**Which procedure is used to visualize blood vessels and may involve contrast dye and catheterization?**

✓ **Angiography**

Ultrasound

MRI

PET

2. **ANALYZE THE CASE**

**A doctor threads a thin catheter into an artery and injects dye to look for a blocked vessel. This procedure is:**

✓ **Angiography**

Nuclear scan

Ultrasound

X-ray of the bone

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**Explanation shown if missed:** Angiography is used to see BLOOD VESSELS. It may use a contrast dye and a thin tube (catheter) placed in a vessel.